Question	Answer	Marks	Partial Marks
5(a)(v)	[y =] $\frac{1}{3}x$	**4**	**B3** for a correct equation in the wrong form as final answer Or **B2** for 1/3 stated or used as perpendicular gradient OR **M1** for [grad PQ] = $\frac{7--5}{1-5}$ oe **M1** for $\frac{-1}{\textit{their}\ \text{grad}\ PQ}$ **M1dep** for substituting *their***(a)(iv)** or (0,0) into y = *their* $mx + c$ oe dep on the 2nd M1 or B2
5(b)	$\frac{3}{5}$ **a** + $\frac{2}{5}$ **b** final answer	**4**	**B3** for an unsimplified correct answer or **B2** for $AM = \frac{2}{5}(\boldsymbol{b}-\boldsymbol{a})$ soi or $BM = \frac{3}{5}(\boldsymbol{a}-\boldsymbol{b})$ soi or **B1** for AB = **b – a** or BA = **a – b** or for a correct route for OM or for correct diagram
6(a)	245	**1**	
6(b)(i)	180 – (55 + 25) [=100]	**M1**	
6(b)(ii)	$\frac{32\times\sin 25}{\sin 100}$ oe	**M2**	**M1** for $\frac{\sin 25}{BH} = \frac{\sin 100}{32}$ oe
	13.73…	**A1**	

Question	Answer	Marks	Partial Marks
6(c)	258 or 257.9 to 258.0…	**5**	**B4** for 67.9 to 68.0… OR **M2** for [cos =] $\left(\frac{11^2+13.7^2-14^2}{2\times13.7\times11}\right)$ **A1** for 0.3738 to 0.376 or **M1** for $14^2 = 11^2 + 13.7^2 - 2\times 11 \times 13.7 \times \cos B$ **M1dep** on at least M1 for 190 + *their* angle *B*
6(d)(i)	2 44 pm or 14 44 cao	**4**	**B3** for 1 hour 44 or 1 hour 43.6 to 1 hour 43.8 or 104 or 103.6 to 103.8 or **B2** for 1.727 to 1.73 or **M2** for $\frac{32}{10\times1.852}\times60$ or **M1** for 32 ÷ (10 × 1.852)
6(d)(ii)	7.857 to 7.88	**3**	**M2** for $\frac{x}{13.7}=\cos55$ oe or **M1** for dist to *H* occurs when perpendicular from *B* meets *CH* soi
7(a)(i)	10 100	**3**	**M2** for $30\times70+2\times30\times40+2\times40\times70$ or **M1** for 30 × 40 or 30 × 70 or 40 × 70
7(a)(ii)	16	**3**	**M2** for 2 fit width, 2 fit height and 4 fit length soi or **M1** for 70, 30 or 40 ÷ 15 or 20
7(b)(i)	$\frac{1}{3}\pi r^2\times3r$ = *their* (750 ÷ 8.9) oe	**M2**	**M1** for using 750 and 8.9 correctly in $v=m/d$ oe or 750 ÷ 8.9
	$r^3=\frac{their(750\div8.9)}{\pi}$ oe	**M1dep**	
	$r = 2.993\ldots$	**A1**	